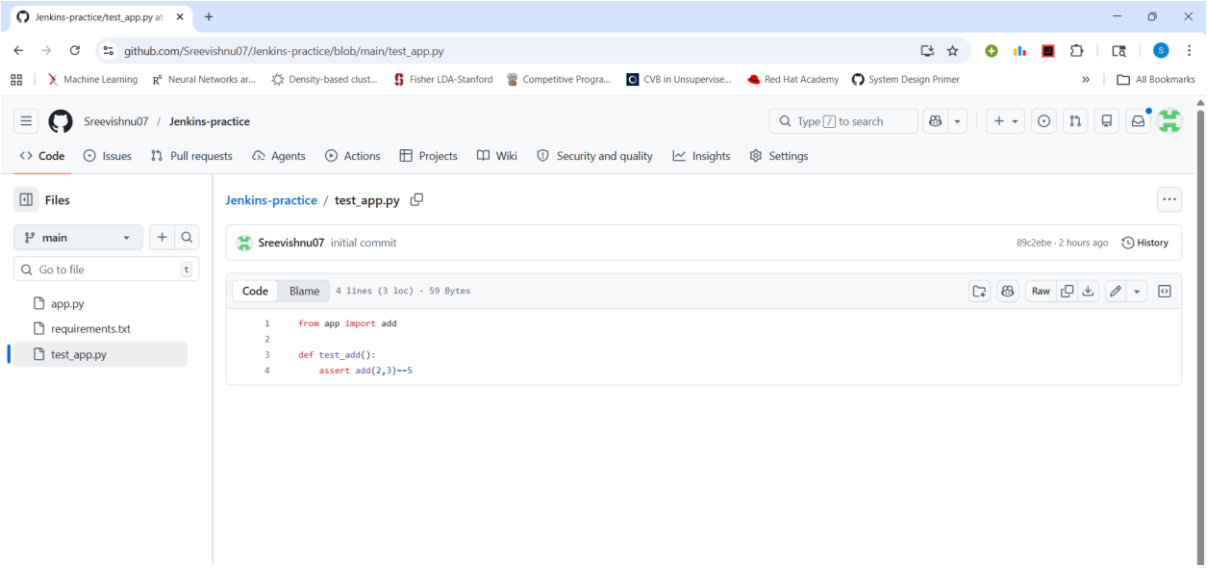
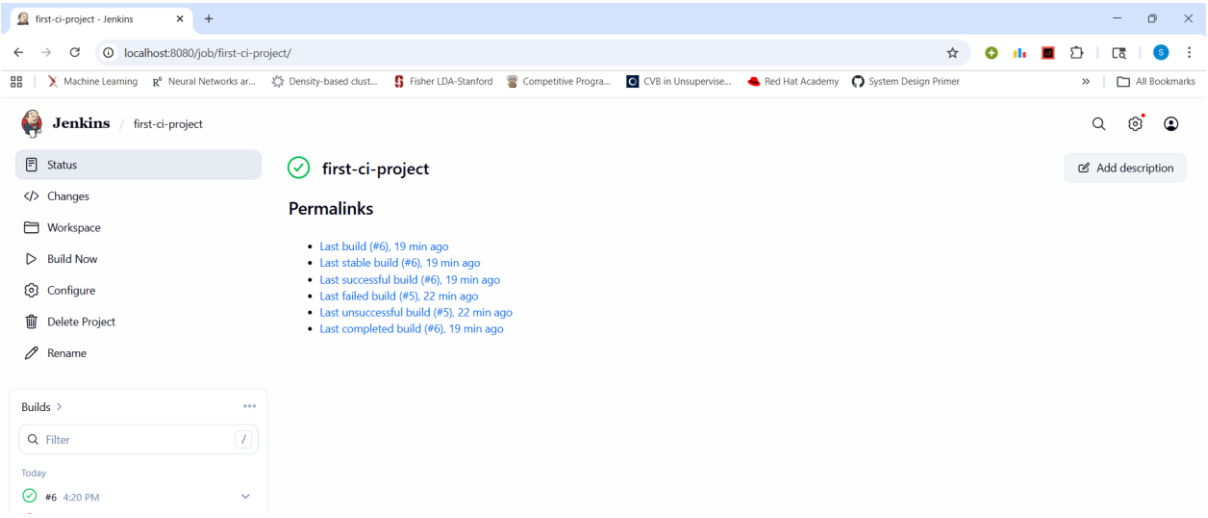
Complete Jenkins Mastery(Absolute Fav)

1. Implementation of a Continuous Integration (CI) Pipeline using Jenkins for Python Testing





roject/configure

Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmarks

nfigure

Git ?

Repositories ?

Repository URL ?

https://github.com/Sreevishnu07/Jenkins-practice.git

Credentials ?

- none -

+ Add

Advanced

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

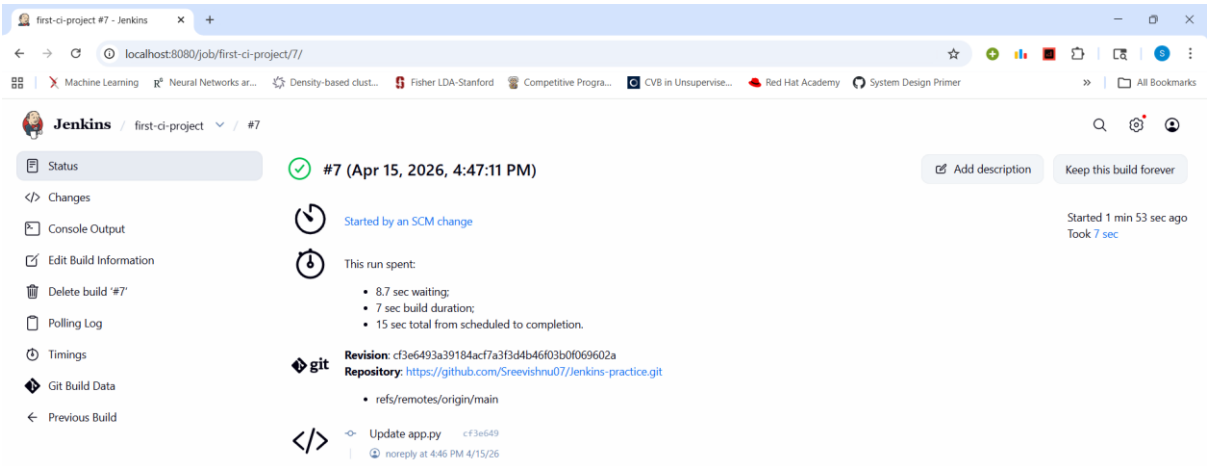
```
python -m pip install -r requirements.txt
python -m pytest
```

Advanced

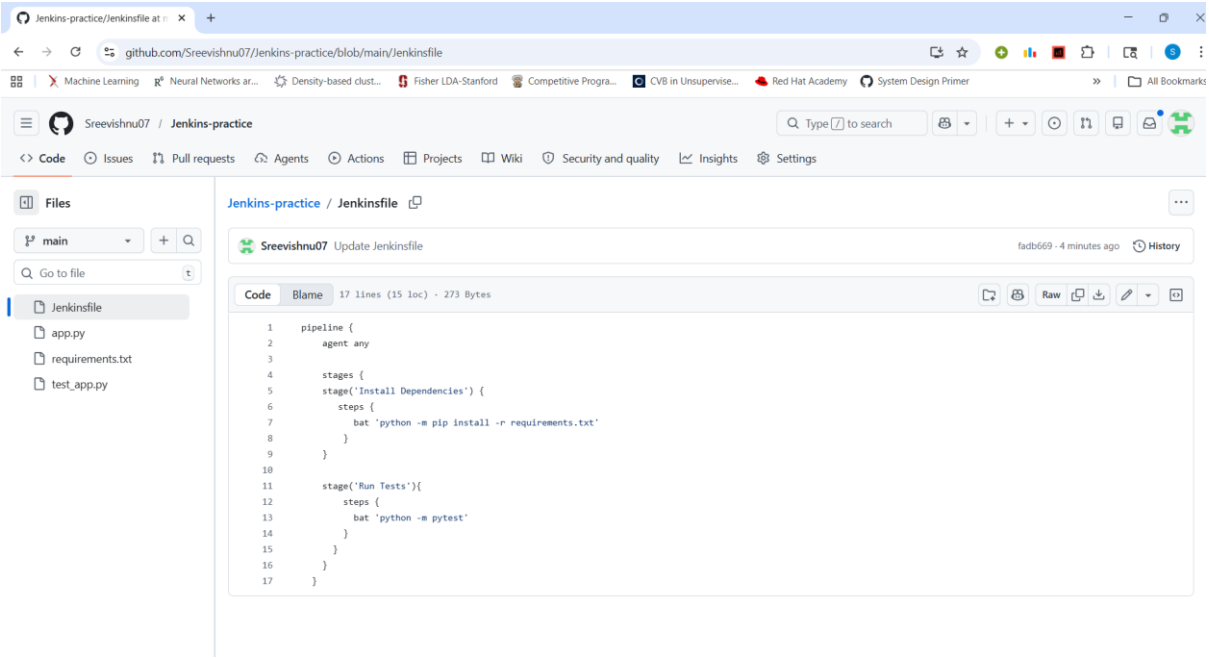
+ Add build step

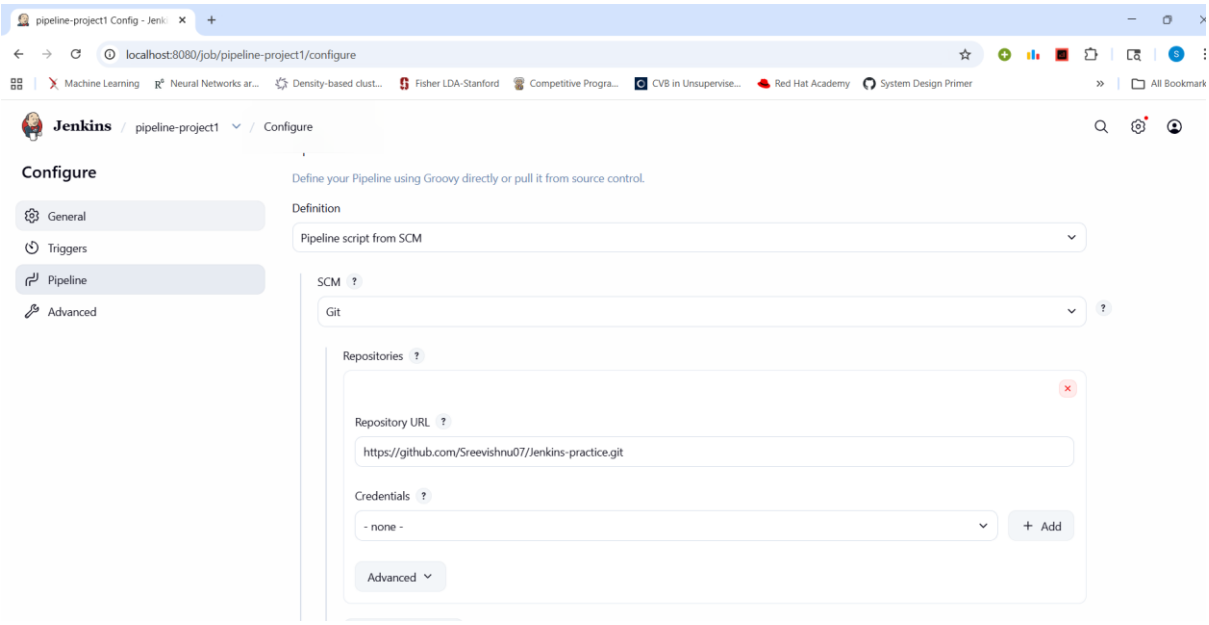
Save | Apply

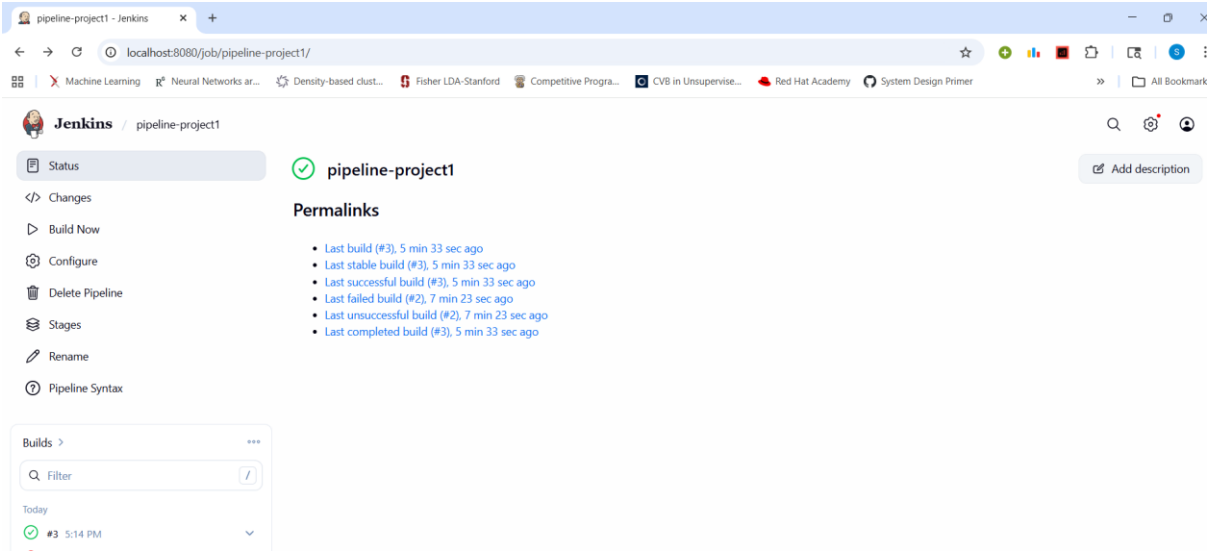
2. Implementation of Automatic Build Trigger using Poll SCM



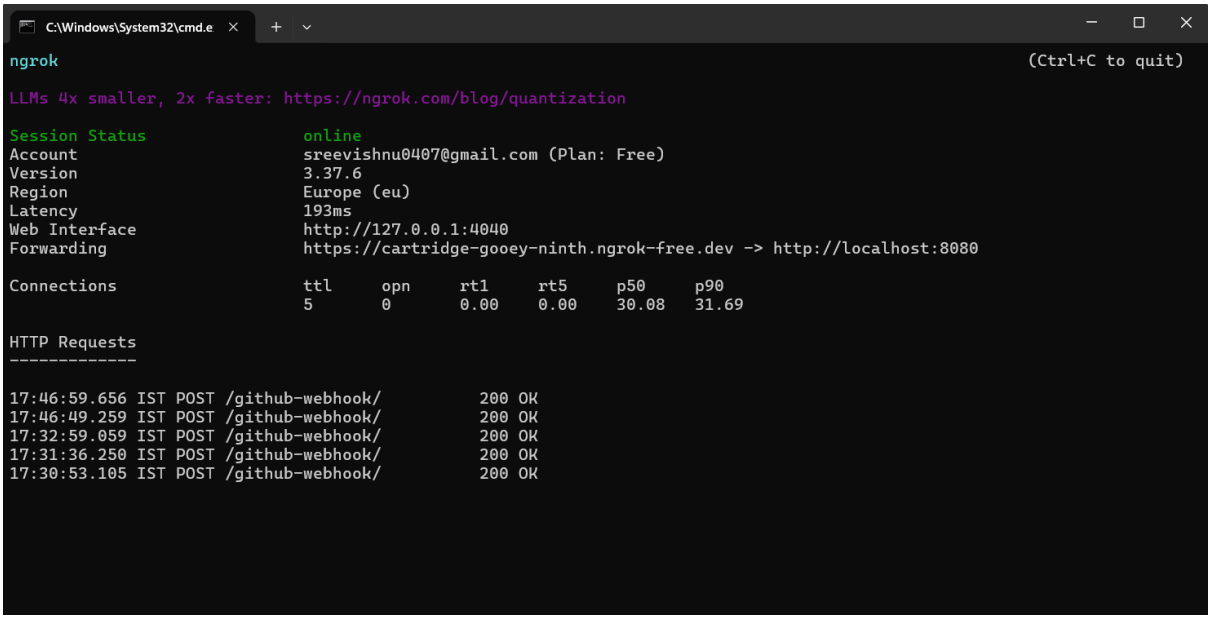
3. Implementation of CI Pipeline using Jenkinsfile with Pipeline Script from SCM

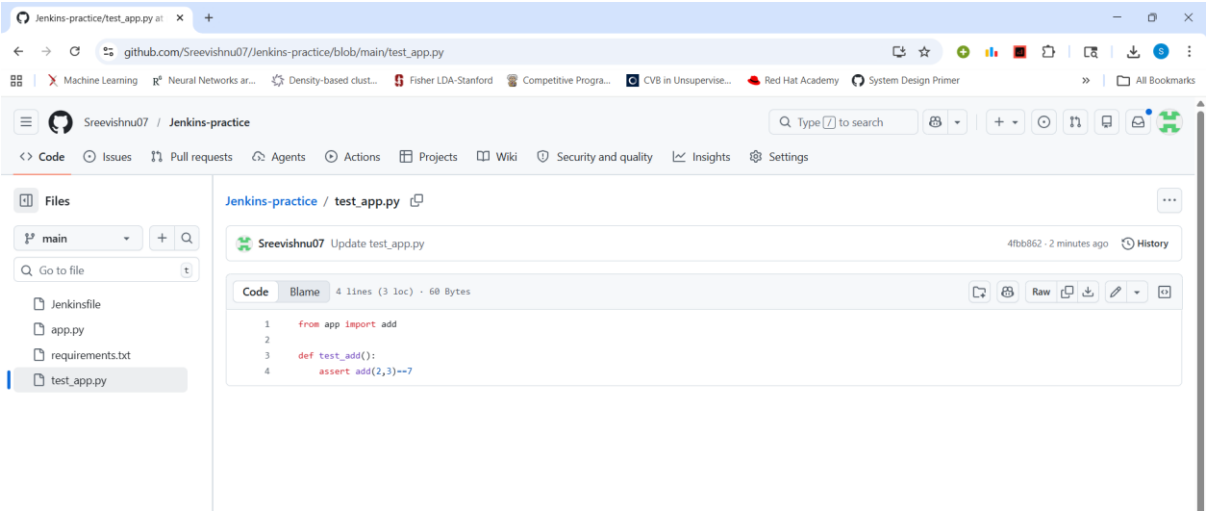

Jenkins-practice/Jenkinsfile at
github.com/Sreevishnu07/Jenkins-practice/blob/main/Jenkinsfile
Sreevishnu07 / Jenkins-practice
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
Jenkinsfile
app.py
requirements.txt
test_app.py
Jenkins-practice / Jenkinsfile
Sreevishnu07 Update Jenkinsfile
fadb669 · 4 minutes ago
History
Code
Blame
17 lines (15 loc) · 273 Bytes
Raw
pipeline {
agent any
stages {
stage('Install Dependencies') {
steps {
bat 'python -m pip install -r requirements.txt'
}
}
stage('Run Tests'){
steps {
bat 'python -m pytest'
}
}
}
}

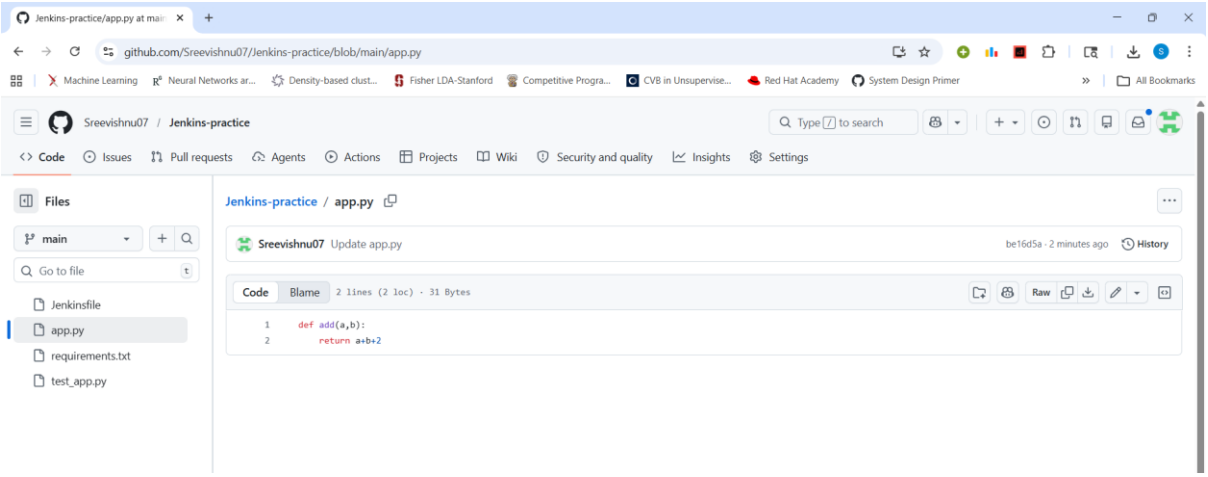

pipeline-project1 Config - Jenk
localhost:8080/job/pipeline-project1/configure
Jenkins / pipeline-project1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script from SCM
SCM
Git
Repositories
Repository URL
https://github.com/Sreevishnu07/Jenkins-practice.git
Credentials
- none -
Add
Advanced

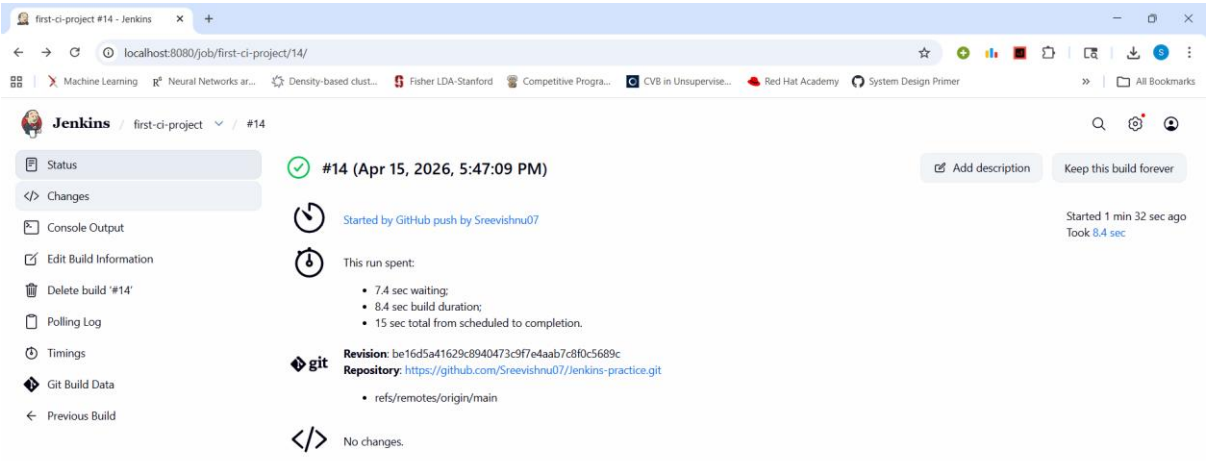


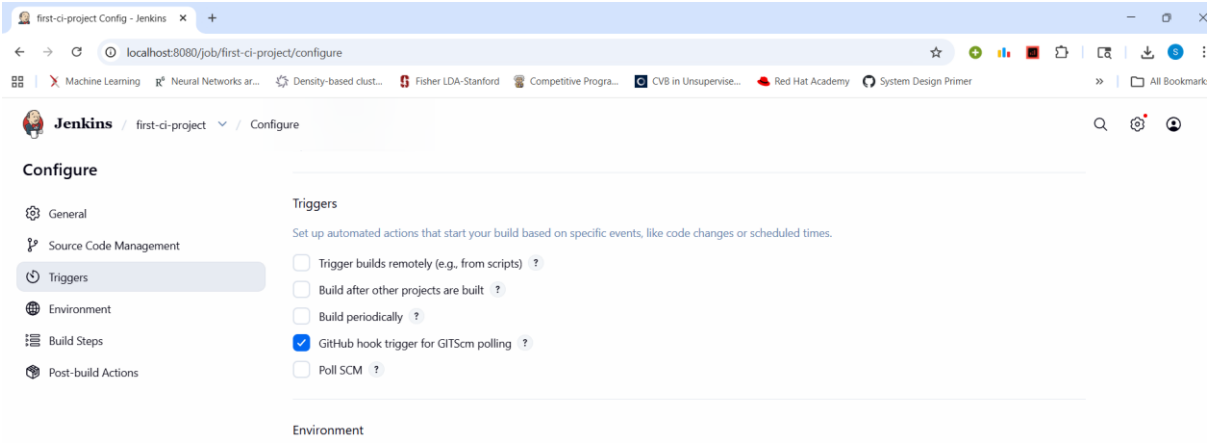
4. CI Pipeline with GitHub Webhook Integration using Jenkins and ngrok



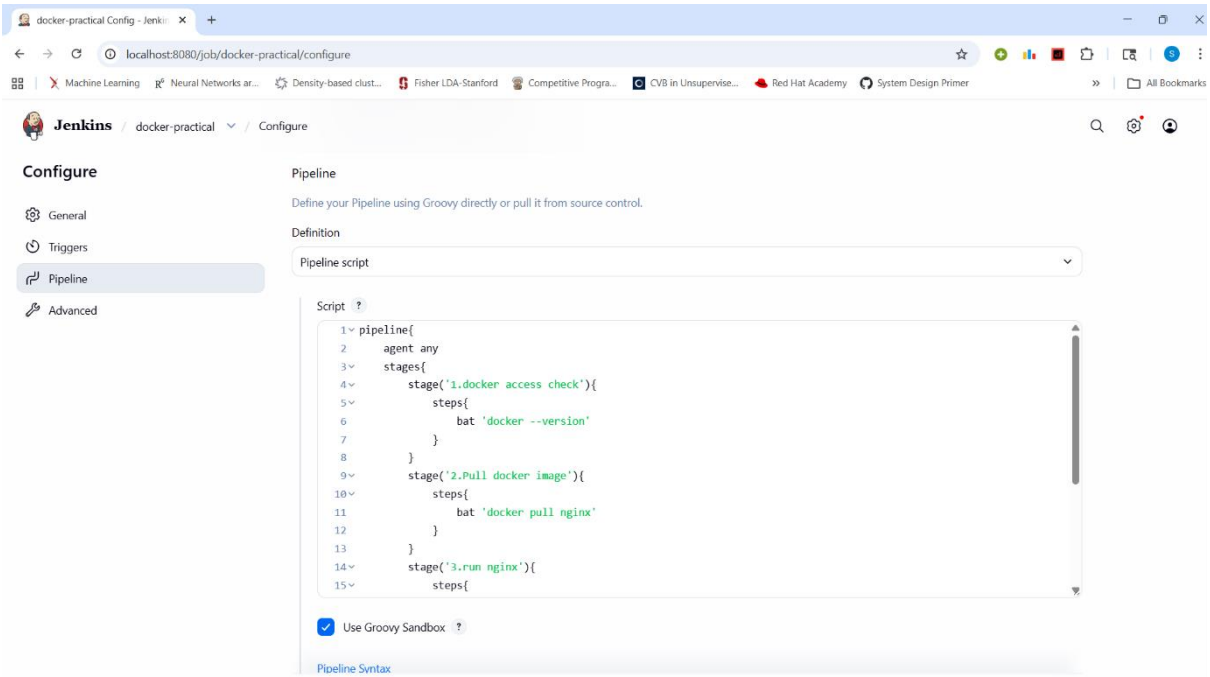

Jenkins-practice/test_app.py at
github.com/Sreevishnu07/Jenkins-practice/blob/main/test_app.py
Sreevishnu07 / Jenkins-practice
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
Files
main
Go to file
Jenkinsfile
app.py
requirements.txt
test_app.py
Jenkins-practice / test_app.py
Sreevishnu07 Update test_app.py
4fbb862 · 2 minutes ago History
Code Blame 4 lines (3 loc) · 60 Bytes
from app import add
def test_add():
assert add(2,3)==7

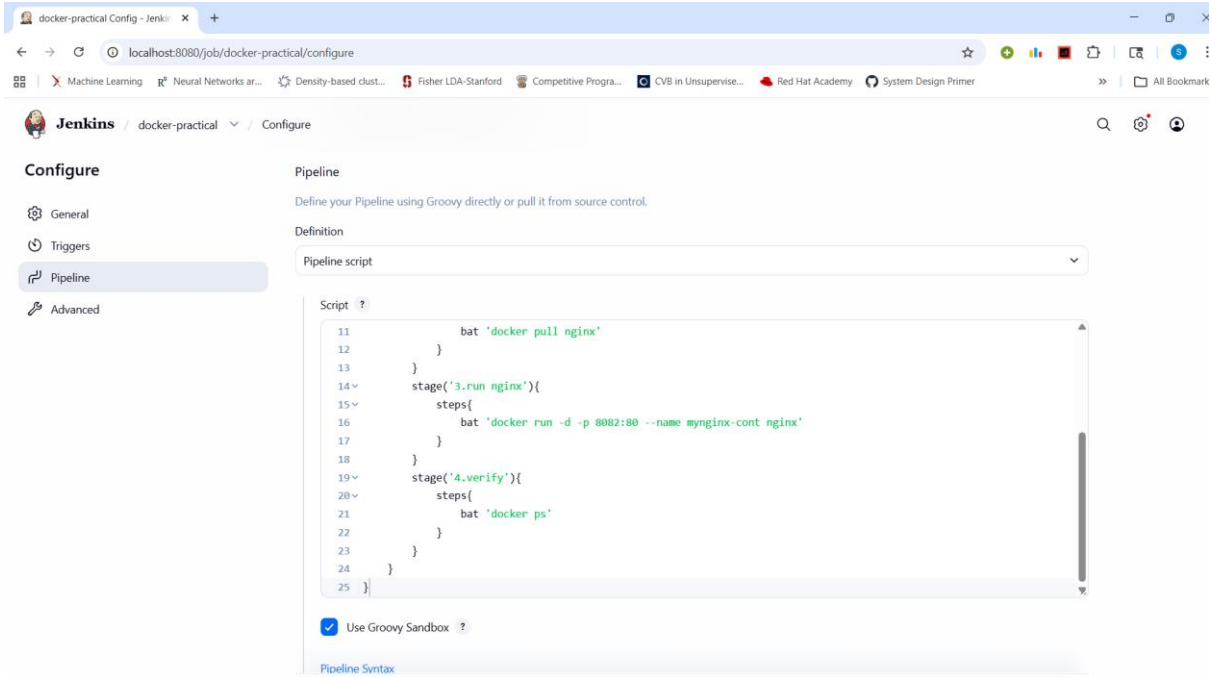

Jenkins-practice/app.py at main
github.com/Sreevishnu07/Jenkins-practice/blob/main/app.py
Sreevishnu07 / Jenkins-practice
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
Files
main
Go to file
Jenkinsfile
app.py
requirements.txt
test_app.py
Jenkins-practice / app.py
Sreevishnu07 Update app.py
be16d5a · 2 minutes ago History
Code Blame 2 lines (2 loc) · 31 Bytes
def add(a,b):
return a+b+2

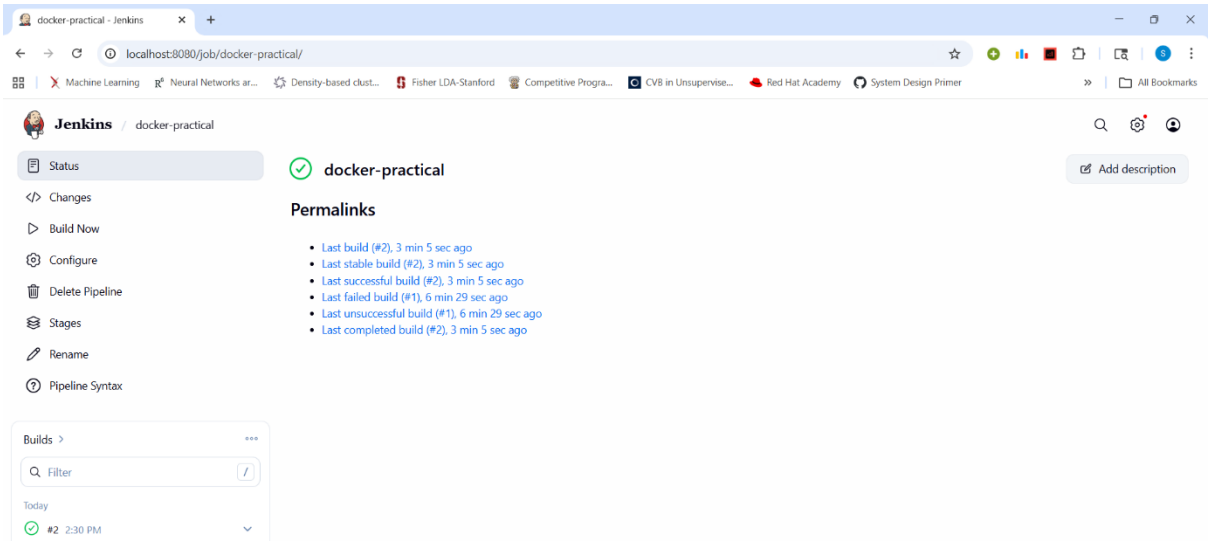

first-ci-project #14 - Jenkins
localhost:8080/job/first-ci-project/14/
Jenkins / first-ci-project / #14
Status
Changes
Console Output
Edit Build Information
Delete build '#14'
Polling Log
Timings
Git Build Data
Previous Build
#14 (Apr 15, 2026, 5:47:09 PM)
Add description
Keep this build forever
Started by GitHub push by Sreevishnu07
Started 1 min 32 sec ago
Took 8.4 sec
This run spent:
7.4 sec waiting;
8.4 sec build duration;
15 sec total from scheduled to completion.
Revision: be16d5a41629c8940473c9f7e4aab7c8f0c5689c
Repository: https://github.com/Sreevishnu07/Jenkins-practice.git
refs/remotes/origin/main
No changes.

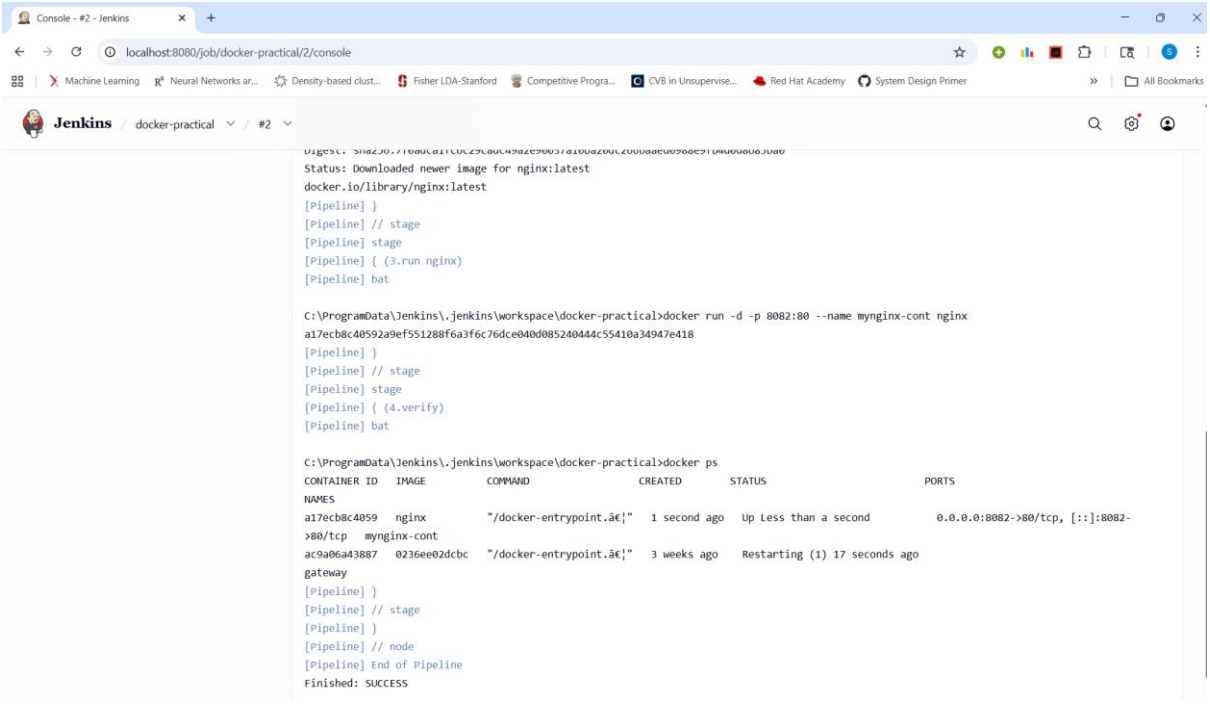


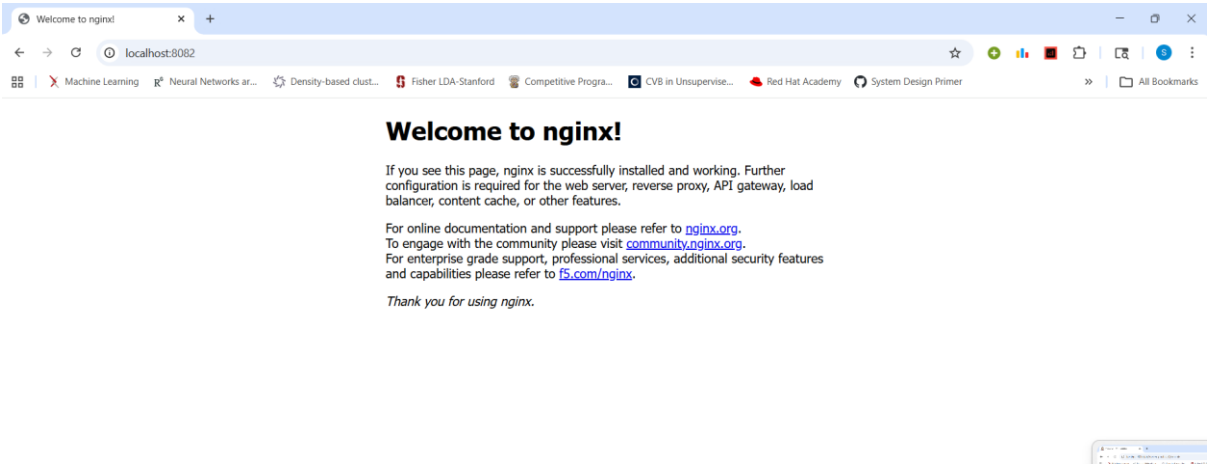
5. Automated Container Deployment using Jenkins and Docker (NGINX)




docker-practical Config - Jenki
localhost:8080/job/docker-practical/configure
Jenkins / docker-practical / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
bat 'docker pull nginx'
}
}
stage('3.run nginx'){
steps{
bat 'docker run -d -p 8082:80 --name mynginx-cont nginx'
}
}
stage('4.verify'){
steps{
bat 'docker ps'
}
}
}
}
Use Groovy Sandbox
Pipeline Syntax


docker-practical - Jenkins
localhost:8080/job/docker-practical/
Jenkins / docker-practical
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
#2 2:30 PM
docker-practical
Add description
Permalinks
Last build (#2), 3 min 5 sec ago
Last stable build (#2), 3 min 5 sec ago
Last successful build (#2), 3 min 5 sec ago
Last failed build (#1), 6 min 29 sec ago
Last unsuccessful build (#1), 6 min 29 sec ago
Last completed build (#2), 3 min 5 sec ago





6. CI Pipeline for Java Application Build using Jenkins and Maven

Definition

Pipeline script

Script

```
pipeline{
    agent any

    tools{
        maven 'Maven-3'
    }

    stages{
        stage('1.maven access check'){
            steps{
                bat 'mvn --version'
            }
        }
        stage('2.clean workspace'){
            steps{
```

Use Groovy Sandbox

Pipeline Syntax

Definition

Pipeline script

Script

```
        stage('3.archetype generate'){
            steps{
                bat '''
                mvn archetype:generate -DgroupId=com.example ^
                -DartifactId=my-app ^
                -DarchetypeArtifactId=maven-archetype-quickstart ^
                -DinteractiveMode=false
                '''
            }
        }
        stage('4.build project'){
            steps{
                bat 'cd my-app && mvn clean package'
            }
        }
```

Use Groovy Sandbox

Pipeline Syntax

Definition

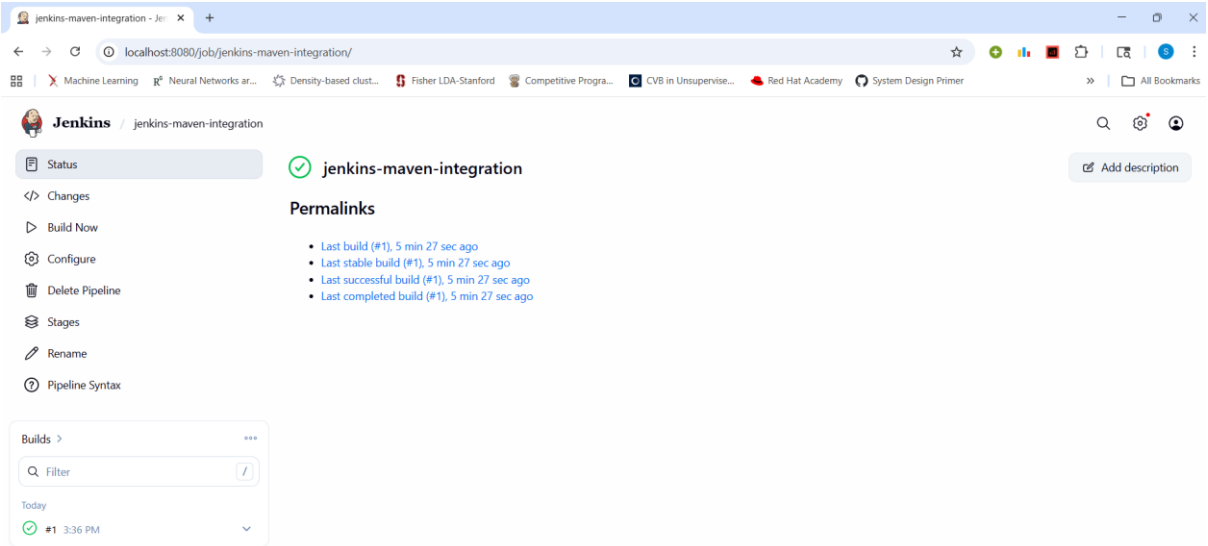
Pipeline script

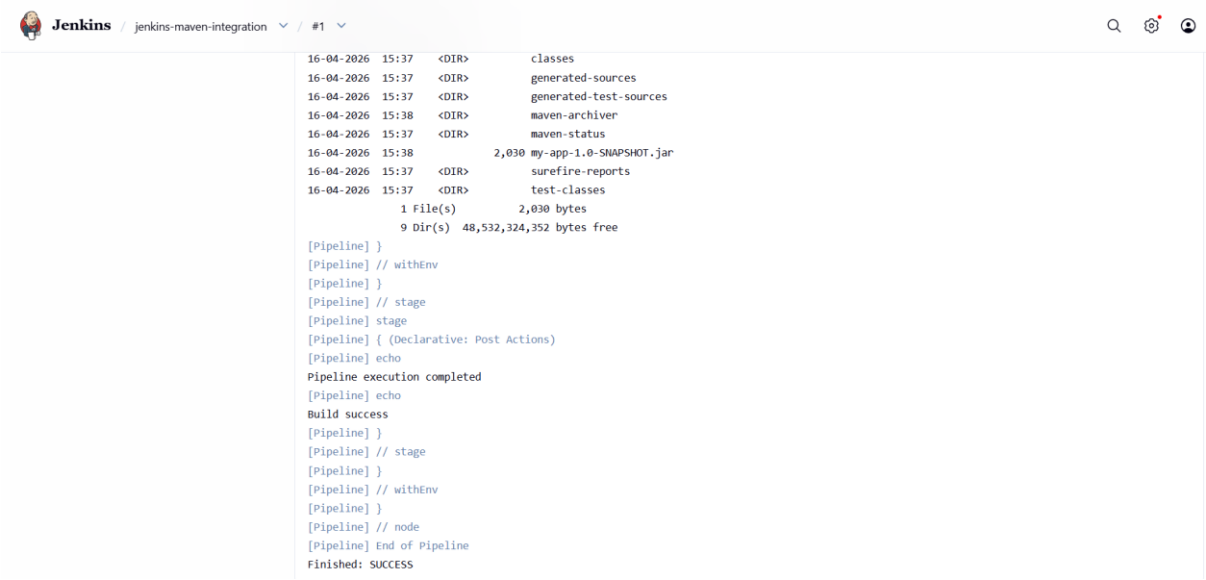
Script ?

```
        stage('5.verify build'){
            steps{
                bat 'dir my-app\\target'
            }
        }
    }
    post{
        always{
            echo 'Pipeline execution completed'
        }
        success{
            echo 'Build success'
        }
        failure{
            echo 'Build Failure'
```

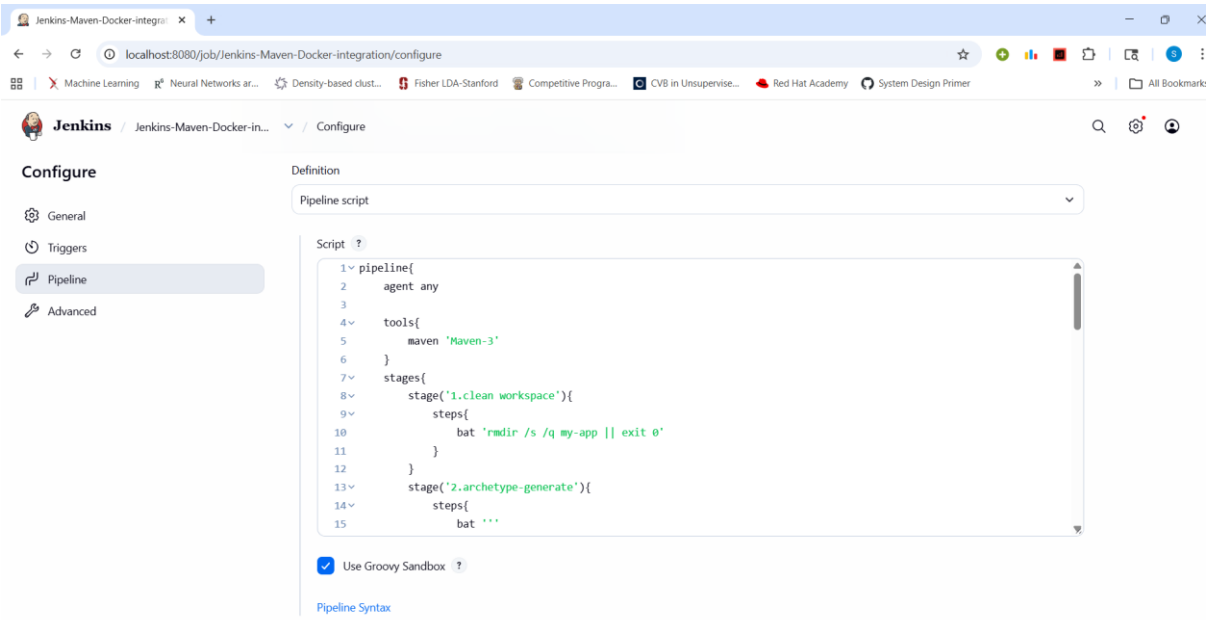
Use Groovy Sandbox ?

Pipeline Syntax





7. CI/CD Pipeline for Java Application using Jenkins, Maven Build, and Docker Containerization



Definition

Pipeline script

Script ?

```
                bat '''
                mvn archetype:generate ^
                -DgroupId=com.example ^
                -DartifactId=my-app ^
                -DarchetypeArtifactId=maven-archetype-quickstart ^
                -DinteractiveMode=false
                '''
            }
        }
        stage('3.Build JAR'){
            steps{
                bat 'cd my-app && mvn clean package'
            }
        }
        stage('4.Create Dockerfile'){
```

Use Groovy Sandbox ?

Pipeline Syntax

Script ?

```
            steps{
                writeFile file:'my-app/Dockerfile',text: '''
                FROM eclipse-temurin:17-jdk
                WORKDIR /app
                COPY target/my-app-1.0-SNAPSHOT.jar app.jar
                CMD ["java","-cp","app.jar","com.example.App"]
                '''
            }
        }
        stage('5.Build docker image'){
            steps{
                bat 'cd my-app && docker build -t my-java-app .'
            }
        }
        stage('Run Container'){
```

Use Groovy Sandbox ?

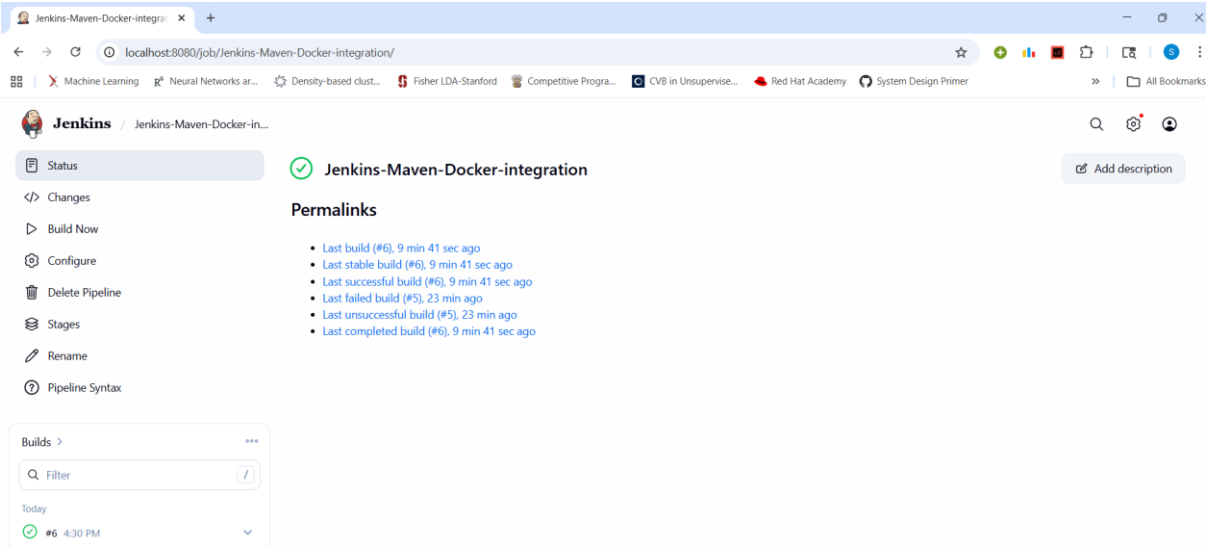
Pipeline Syntax

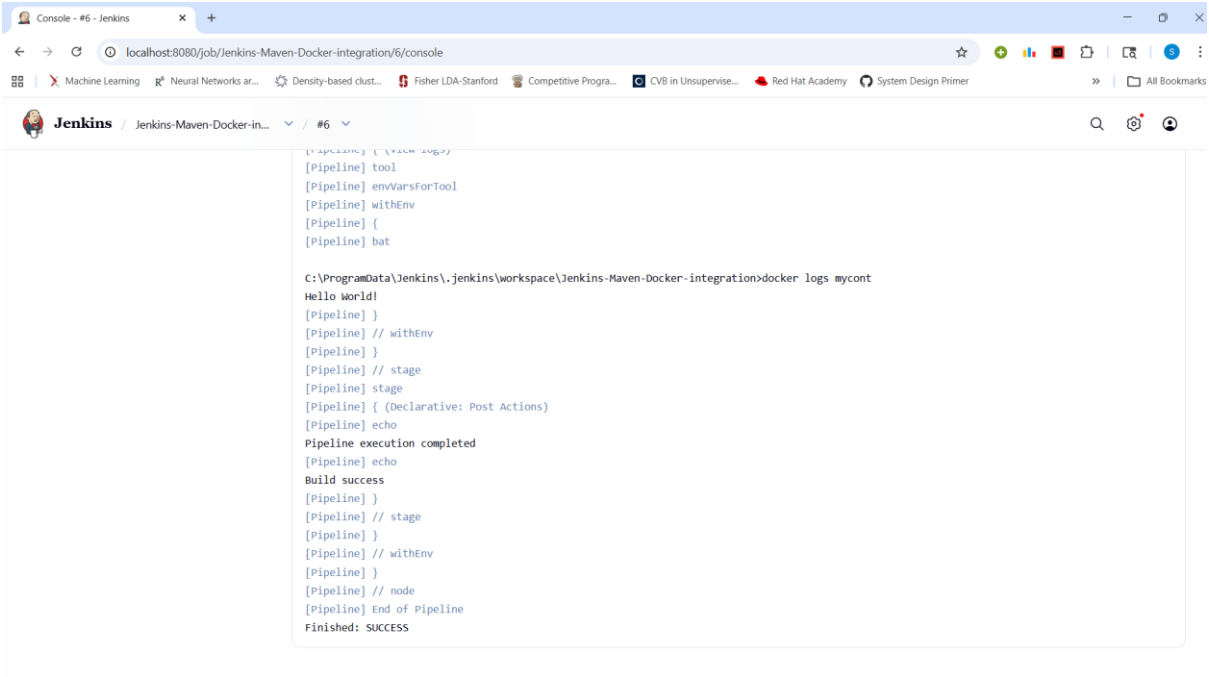
Script ?

```
            steps{
                bat 'docker run -d --name mycont my-java-app'
            }
        }
        stage('view logs'){
            steps{
                bat 'docker logs mycont'
            }
        }
    }
    post{
        always{
            echo 'Pipeline execution completed'
        }
        success{
```

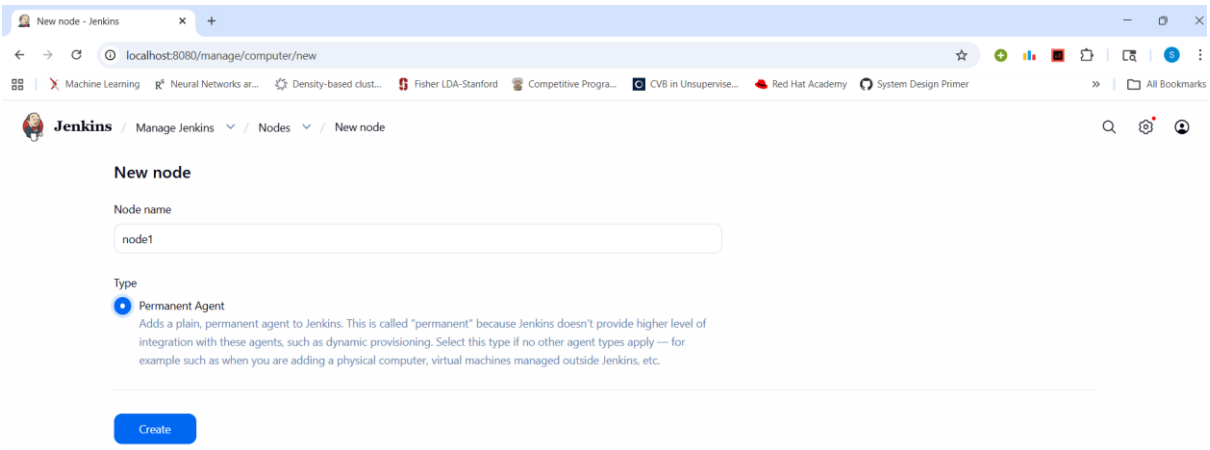
☑ Use Groovy Sandbox ?

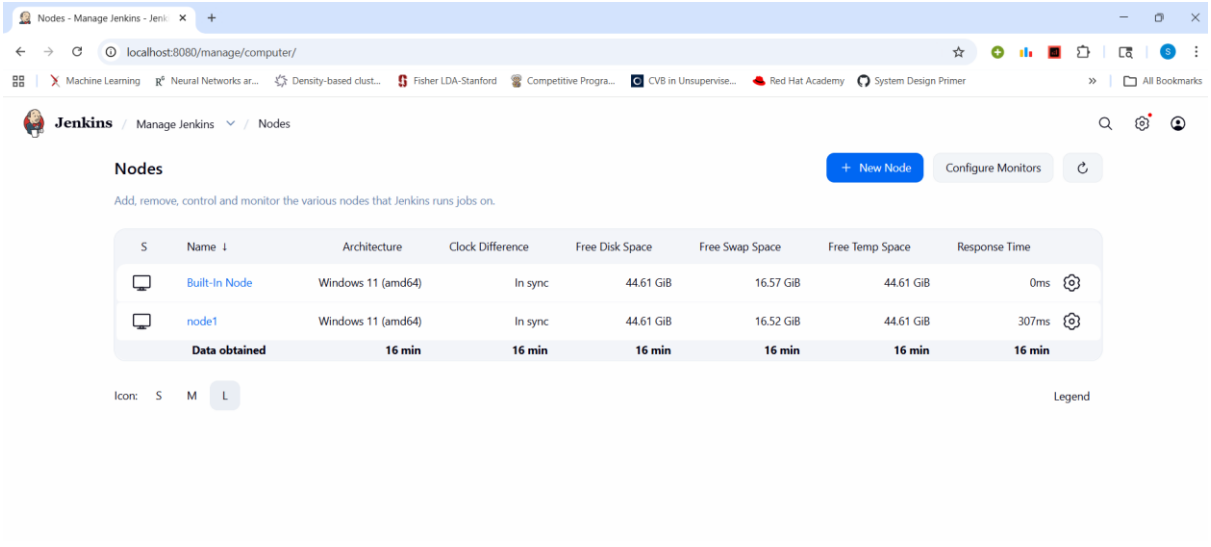
Pipeline Syntax



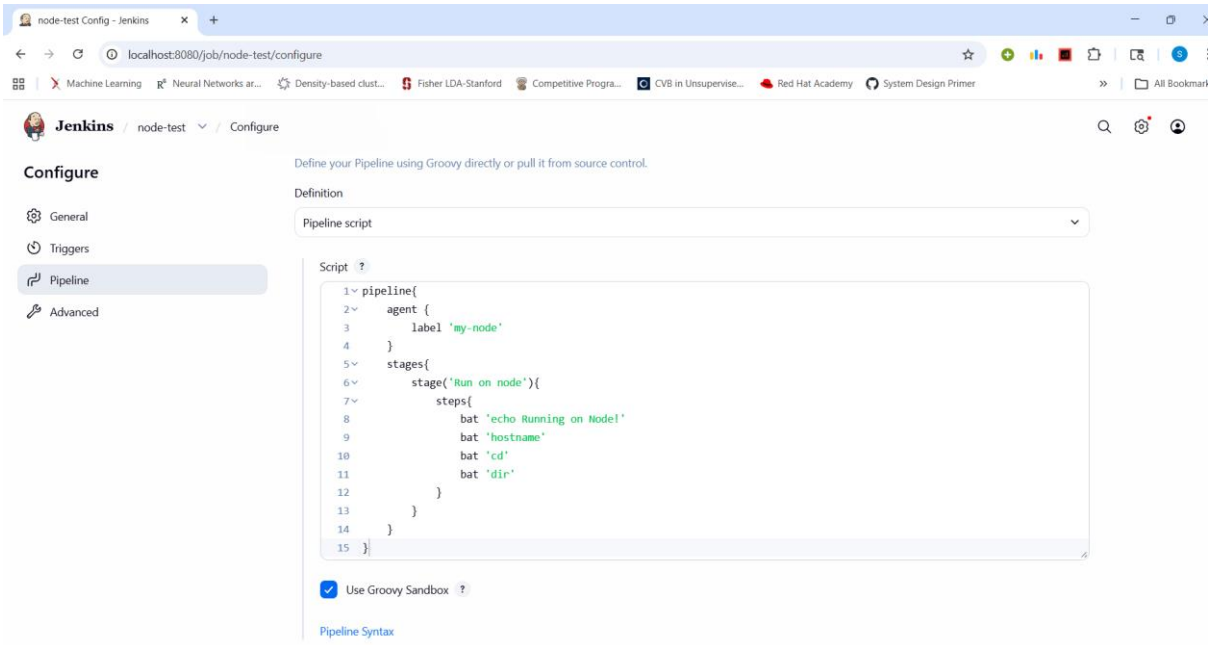


8. Distributed CI Execution using Jenkins Controller-Agent Architecture (Jenkins Distributed Build Execution using Agent Nodes)






```
12936868f19ad861 -name node1 -webSocket -workDir "C:\jenkins-node"
Apr 16, 2026 5:06:33 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\jenkins-node\remoting as a remoting work directory
Apr 16, 2026 5:06:33 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to C:\jenkins-node\remoting
Apr 16, 2026 5:06:33 PM hudson.remoting.Launcher createEngine
INFO: Setting up agent: node1
Apr 16, 2026 5:06:33 PM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3352.v17a_fb_4b_2773f
Apr 16, 2026 5:06:33 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\jenkins-node\remoting as a remoting work directory
Apr 16, 2026 5:06:34 PM hudson.remoting.Launcher$CuiListener status
INFO: WebSocket connection open
Apr 16, 2026 5:06:34 PM hudson.remoting.Launcher$CuiListener status
INFO: Connected
Apr 16, 2026 5:12:10 PM org.jvnet.winp.Native loadByUrl
WARNING: DLL and EXE are inconsistently present on disk
```



node-test - Jenkins

localhost:8080/job/node-test/

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmarks

Jenkins / node-test

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Stages
- Rename
- Pipeline Syntax

Builds

Filter /

Today

#1 5:12 PM

node-test

Add description

Permalinks

- Last build (#1), 13 min ago
- Last stable build (#1), 13 min ago
- Last successful build (#1), 13 min ago
- Last completed build (#1), 13 min ago

Console - #1 - Jenkins

localhost:8080/job/node-test/1/console

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmarks

Jenkins / node-test / #1

- Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Timings
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

```
Started by user Sree Vishnu
[Pipeline] Start of Pipeline
[Pipeline] node
Running on node1 in C:\jenkins-node\workspace\node-test
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Run on node)
[Pipeline] bat

C:\jenkins-node\workspace\node-test>echo Running on Node!
Running on Node!
[Pipeline] bat

C:\jenkins-node\workspace\node-test>hostname
admin
[Pipeline] bat

C:\jenkins-node\workspace\node-test>cd
C:\jenkins-node\workspace\node-test
[Pipeline] bat

C:\jenkins-node\workspace\node-test>dir
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\jenkins-node\workspace\node-test

16-04-2026  17:12    <DIR>          .
16-04-2026  17:12    <DIR>          ..
               0 File(s)              0 bytes
```

```
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\jenkins-node\workspace\node-test

16-04-2026  17:12    <DIR>          .
16-04-2026  17:12    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  47,894,786,048 bytes free
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```